

Week 1

Research Questions, Hypotheses, and Theory

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Check ins

Let's chat about where you are at with finding an advisor, developing a research project, and annotated bibliographies

Let's talk literature

- ▶ We focused on how to look for literature
- ▶ Let's now focus on how to read through literature
- ▶ Grab an article and follow along

Abstracts

- ▶ Is this relevant?
 - ▶ No → move on
 - ▶ Yes → save to Zotero

Introduction

- ▶ New topic = read
 - ▶ Citations may be helpful for developing research idea
 - ▶ Know about the topic → ignore/skim
 - ▶ Look at the final paragraph in this section → RQs/hypotheses/aims/purpose/etc.
- ▶ Contains theory

Methods

- ▶ Very important (expect to add a lot here in annotated bibliographies)
- ▶ Contains protocol for data collection
- ▶ Cleaning methods
- ▶ Description of data
- ▶ Analytic Technique
- ▶ What the researchers did

Results

- ▶ Main findings of the study
 - ▶ Dense
- ▶ I usually read it to get specifics
 - ▶ Actual estimation values

Discussion/Conclusion

- ▶ Reports findings in clear terms
 - ▶ If something is interesting here THEN go to the results to understand more
- ▶ I write out main findings of the study in my annotated bibliographies

Future Studies/Limitations

- ▶ Can find inspiration of how you might be able to expand on research that has been completed
- ▶ You can also take some of the future direction(s) and apply it to your work
 - ▶ If relevant

Example Article

- ▶ Exploring Approaches for Estimating Parameters in Cognitive Diagnosis Models with Small Sample Sizes
- ▶ Use your own article if you want

If you haven't already, get an article

Let's spend 10 minutes reading your article and filling in some information for your annotated bibliography

JP's Article - Intro

▶ Intro

- ▶ Grab articles on estimation (section 1.2)
- ▶ Purpose/Goal: “this study redounds to the line of work on the use of CDM in small samples [38] with the aim of concluding the best way to estimate item parameters”
- ▶ Hypothesis: better performance of the different alternatives against MMLE-EM in situations of low sample size and no differences in situations of large sample size.
 - ▶ Hypothesis 1a: better performance of the different alternatives against MMLE-EM in situations of low sample size
 - ▶ Hypothesis 1b: no differences of different alternatives against MMLE-EM in situations of large sample size

Methods

- ▶ Gold standard library for DCMs (GDINA) for MMLE-EM
 - ▶ MCMC using JAGS; compared to other acceptable programs
- ▶ Compared priors (look into priors for my work)
- ▶ More information on simulation study (should implement myself)
 - ▶ samples of 100 & 2000
 - ▶ 30 items & 5 skills
- ▶ Used well-known assessment dataset (not really empirical study; limitation to expand on in own work)

Results

- ▶ Didn't bother reading yet

Discussion

- ▶ MCMC performed better than MMLE-EM
- ▶ Restricted model did well on classification accuracy, reliability
 - ▶ Be cautious of using this model because it is very restrictive
 - ▶ Always compare to non-restricted model
- ▶ When working with larger data ($N \geq 1000$), use MMLE-EM because it is faster and as accurate as MCMC
- ▶ Conclusion: For small samples use MCMC and/or restricted model

Icebreaker

From the literature you have read, what is a potential research question you are thinking about? Draft a research question to share with your group.

Research Questions (RQs), Hypotheses, Aims, Purpose???

- ▶ What your plan is for your research study
- ▶ RQs and hypotheses should be written before conducting analyses
 - ▶ This is good science
 - ▶ Good science $\neq$ science
- ▶ Narrows your purpose statement
- ▶ Having RQs and/or hypotheses = good science
 - ▶ Should be testable

Research Questions (RQs), Hypotheses, Aims, Purpose???

- ▶ RQs ask questions that you want answered
- ▶ Hypotheses are statements of what you hope to find
- ▶ Aims are exploratory ways of stating what you are interested in
- ▶ Purpose statements are what is the overarching reason you are conducting this study

Example RQs

- ▶ Does student math self-efficacy improve by the end of the semester?
- ▶ What courses determine if students become accepted to be DS honors recipients?
- ▶ How do students learn using alternative course designs?

RQs for Qualitative Analyses

- ▶ RQs are crucial in qualitative research
- ▶ Central questions on the broad phenomenon
 - ▶ Subquestions to narrow the focus of the study
- ▶ Central question should include your strategy for analysis
- ▶ Use “what” or “how” to create an open-ended question

Qualitative RQ Template

(How or What) is the (“story for” for narrative research; “meaning of” the phenomenon for phenomenology; “theory that explains the process of” for grounded theory; “issue” in the “case” for case study) of (phenomenon) for (participants) at (research site)?

RQs for Quantitative (Inferential) Analyses

- ▶ Depends on the model
- ▶ Can focus on specific variables and relationships of interest?
- ▶ Depending on the analysis and design, can include causal language
 - ▶ Affect, cause, impact, etc.
- ▶ Can be more broad and followed by specific hypotheses
- ▶ Theory testing can be broad

RQs for Quantitative Analyses

- ▶ Options include
 - ▶ What are the roles of features a-d in predicting y?
 - ▶ Is x related to y?
 - ▶ Does x cause y when accounting for z?
 - ▶ Are there differences in the relationship between x and y for z (groups)?
 - ▶ Does the theory of _____ support z (1 group)?

RQs for Quantitative (Predictive) Analyses

- ▶ Can also be model specific
- ▶ Are there differences in model fit metrics when imputing data by mean, median, or multiple imputation?
- ▶ Is there a substantial difference in fit metrics comparing model A, model B, etc.?
 - ▶ What is the computational tradeoff of model A compared to model B in predicting _____?
- ▶ Does deleting non-significant features influence model accuracy?

Hypotheses

- ▶ “and” does not belong in a hypothesis EVER!
 - ▶ “and” clearly indicates two parts of a hypothesis or two hypotheses
- ▶ Requirements
 - ▶ Can be tested
 - ▶ Does not include “and”
 - ▶ Wording must match your analysis(es) → affect = causal language
 - ▶ Directionality



Photo by Pixabay

More on RQs & Hypotheses

- ▶ Many times you don't need both
- ▶ Options
 - ▶ Broad RQ & specific hypothesis
 - ▶ Just RQs
 - ▶ Just hypotheses
 - ▶ Null hypothesis (don't declare), but change the language to be RQ
- ▶ Hypotheses are mainly for change
 - ▶ Also called alternative hypotheses
 - ▶ No change/effect = null hypothesis
 - ▶ As scientists we assume there is no change/difference/effect

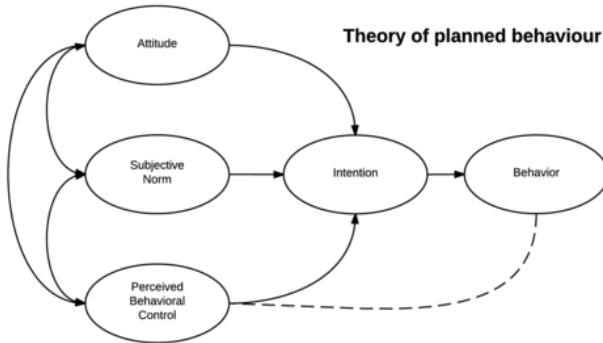
Best Practices

- ▶ Once you state your hypothesis and look at data there is no going back
 - ▶ Funding agencies now require authors to have RQs and/or hypotheses before starting analyses
 - ▶ The reality is that many academics don't follow this rule
 - ▶ Issue with publish or perish reality
 - ▶ No one wants to publish null findings → reproducibility issue

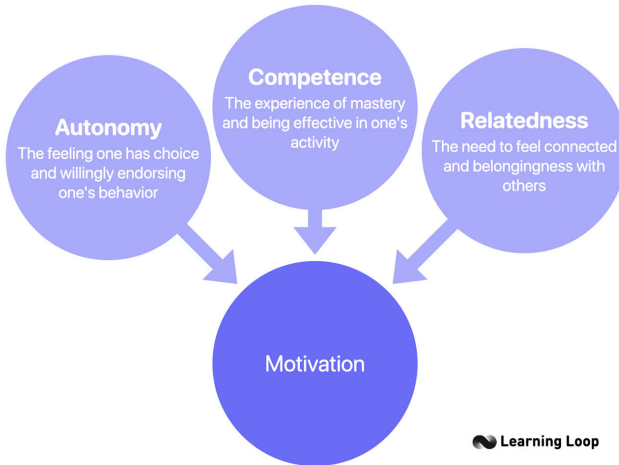
What is a theory/theoretical framework?

- ▶ Guiding your work
- ▶ Find within your literature
- ▶ May want a tab in your annotated bibliographies for theory/framework
- ▶ A large amount of literature supports a specific claim
 - ▶ Meta analyses/systematic reviews
- ▶ May come from different fields
- ▶ Don't make your own theory; you definitely don't have enough background information
- ▶ Don't name a theory after yourself

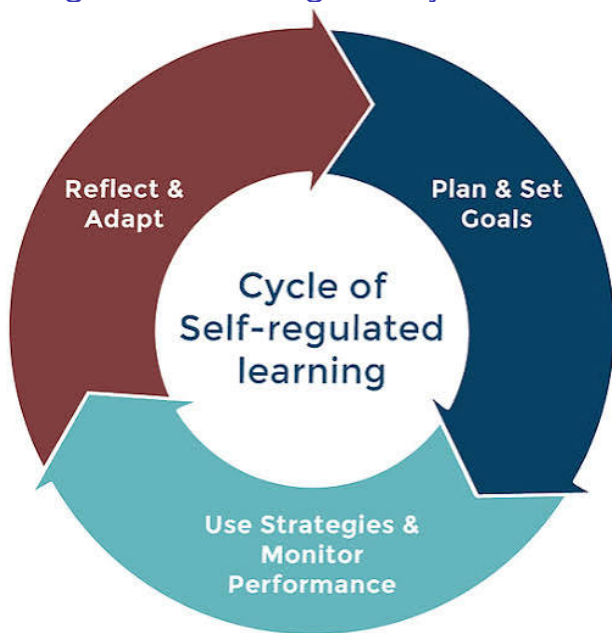
Theory of Planned Behavior



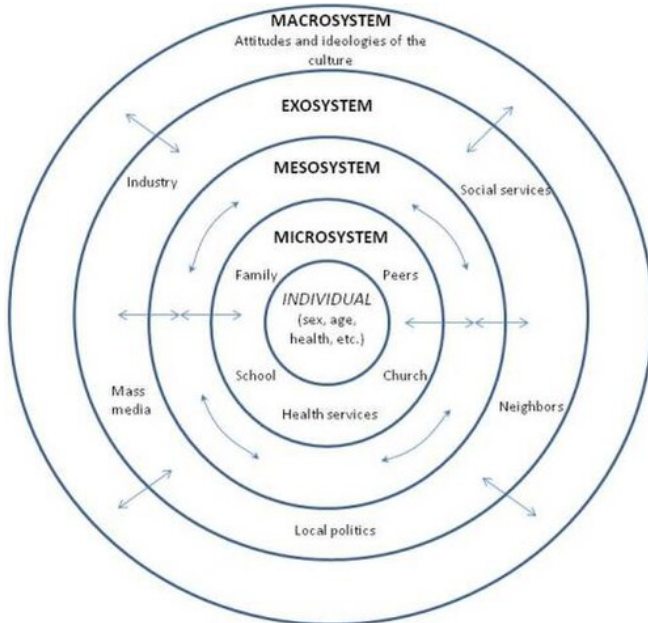
Self-Determination Theory



Self Regulated Learning Theory



Ecological Model



Dynamic Capabilities Theory

